

Samir Amin

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Education

Assam Engineering College, Guwahati, Assam

Expected 2026

Master of Computer Applications (MCA)

Projects

Wearable Health Monitoring System

ESP32-C3, MAX30102, MPU6500, TensorFlow Lite, Reinforcement Learning (DQN), BLE, Android

- Designed a wearable device to collect real-time heart-rate and motion sensor data.
- Processed and transmitted sensor data to a mobile application using Bluetooth Low Energy.
- Applied Reinforcement Learning (Deep Q-Network) to detect abnormal movement and distress patterns.
- Developed an Android application to visualize live data and generate alerts.
- Focused on real-time processing in resource-constrained environments.

Doctor Appointment Booking System

HTML, CSS, JavaScript, Python, LLaMA

- Developed a web application for booking and managing doctor appointments.
- Integrated an AI chatbot to assist users with appointment-related queries.
- Improved user experience by automating basic patient interactions.

Food Fest – Online Food Ordering System

HTML, CSS, JavaScript, PHP, MySQL

- Built a full-stack food ordering platform with user and admin panels.
- Implemented cart functionality and database-driven order management.
- Ensured secure and structured data handling.

AI Snake Game Agent

Python, TensorFlow

- Implemented a Reinforcement Learning agent using Deep Q-Networks.
- Used experience replay and reward design to improve gameplay performance.

Skills

Artificial Intelligence: Reinforcement Learning (DQN), TensorFlow, Keras

Programming: Python, JavaScript, C++

Web Development: HTML, CSS, PHP

Tools: Git, Linux, VS Code, Figma

Languages

English – Professional Proficiency

Hindi – Professional Proficiency

Assamese – Native Proficiency